# Zixun Zhou

zixunzhou56@gmail.com | London, UK | github.com/Laplandog

## EDUCATION

**Goldsmiths, University of London** Sep 2026 – present
*MSc Virtual & Extended Realities*

**Royal College of Art** Sep 2025 – Jul 2026
*Graduate Diploma, Design Pathway*
Completed Units 2–4 with two hardware-software prototypes bridging neuroscience, environmental sensing and interactive design. Targeting IDE (Innovation Design Engineering) programme.

**University of Manchester** Sep 2021 – Jun 2024
*BSc Computer Science (Applied)*
Software Engineering, Data Science, AI & Machine Learning, Operating Systems, Computer Architecture.

**UAL Camberwell College of Arts** Sep 2017 – Jun 2018
*Foundation Diploma — Illustration & Game Design*

## PROJECTS

**Anima — Ecological Emotional Translation Device** RCA U3–U4, Jun 2026

- Built an interactive device that reads 8 environmental sensors (temperature, pH, light, moisture, VOC, sound, EC, humidity) via Arduino, processes them through a CfC neural network (PyTorch + NCP), and outputs real-time emotional states as a breathing WS2812 LED and a mobile web dashboard.
- Designed a "genome" system: three habitat presets (freshwater, indoor air, tomato soil) give the device distinct personalities — the same sensor reading produces different emotional responses depending on which ecosystem it represents.
- Developed a narrative engine that writes a first-person "life diary" from the organism's perspective, triggered by comfort-zone transitions rather than raw data ticks.
- Full-stack: Arduino firmware (C++), Python backend (Flask, PyTorch, serial I/O), front-end PWA, hardware assembly.

**Chladni Skin — EEG-Driven Tactile Neurofeedback Device** RCA U2, Mar 2026

- Designed a closed-loop haptic neurofeedback system for ADHD: a silicone membrane filled with iron powder over an electromagnetic coil array, driven by real-time OpenBCI Cyton brainwave data.
- Engineered three adaptive tactile modes — Focus (20 Hz pulse on palm acupuncture points), Relax (10 s sine-wave sweep across four motors), Alert (3-burst all-motor activation) — switching automatically based on EEG state.
- Confirmed a literature gap: no prior research used deformation-based (non-vibration) haptics for neurofeedback (Fleury et al., 2020).
- Hardware: OpenBCI Cyton + Daisy (16-ch EEG), 3D-printed electrode cap, Arduino, 3 electromagnets (12 V), 4 vibration motors (3.3 V).

## EXPERIENCE

**UX Data Analyst Intern — WPP / VMLY&R** Jun – Aug 2023

- Decomposed 8 analytical modules (80+ metrics) covering user value benchmarking, ride experience, customer segmentation, UK geographic distribution and competitive analysis.
- Produced data visualisations (geo-heatmaps, scatter plots, dashboards) and designed core presentation decks for client pitches.

## SKILLS

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| **Languages** | Python, C/C++ (Arduino), Java, JavaScript, HTML/CSS, SQL |
| **Frameworks** | PyTorch, Flask, Spring Boot, BrainFlow, OpenBCI |
| **Hardware** | Arduino, serial protocols, sensor integration, WS2812 LED, electromagnetic actuators, 3D printing (FDM/SLA) |
| **Design** | Figma, Procreate, Photoshop, physical prototyping, silicone casting, PVC prop fabrication |
| **Languages** | Mandarin (native), English (fluent) |